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WAVE ATTENUATION DUE TO ICE COVER: AN EXPERIMENTAL MODEL IN A WAVE-ICE FLUME

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ABSTRACT

Waves penetrate deep into the ice covered seas, inducing breakup of the ice cover. Concomitantly, the ice cover attenuates the wave energy over distance, so that wave impacts die out eventually. Observations of wave attenuation and concurrent wave-induced breakup in the literature are serendipitous due to difficulties in making measurements in ice covered seas. Hence understanding of wave-ice interactions remain uncertain. Here we present measurements of wave propagation through ice covered waters in the new experimental wave-ice facility at the University of Melbourne. The facility comprises of a 14m long and 0.76m wide flume in a refrigerated chamber, where temperatures can be lowered down to -12 degrees Celsius to generate a continuous ice cover on the water surface. A wave maker, installed at one end, is used to generate regular waves, ranging from gently-sloping to storm-like conditions. Wave attenuation rates are determined from video-camera images of the displacements of markers embedded in the ice cover. The experiments investigated wave propagation through the continuous ice cover, breakup, and propagation through the broken ice cover. Spatial evolution of the breakup and geometrical properties of floes are monitored and correlated with incident wave properties. Wave attenuation over broken ice is investigated and compared against the continuous ice case.

INTRODUCTION

Sea ice acts as a refrigerator for the world. Its bright surface reflects solar heat, which would otherwise be absorbed by the dark ocean it conceals, and the salt it expels during the freezing process drives thermohaline circulation, which transports cold water towards the equator. As a result, sea ice plays a crucial

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role in our climate system.

Waves penetrate deep into the ice-covered ocean and impact the ice cover [1]. Concomitantly, the ice cover attenuates the wave energy over distance, so that wave impacts die out eventually [2]. The most heralded impact is the ability of waves to break up the ice cover into floes with diameters comparable to the prevailing wavelengths. Asplin et al [3] report wave-induced breakup 250 *km* into the Beaufort Sea; Kohout et al [4] observed breakup events over 300 *km* into the Antarctic sea ice. The resulting region covered by broken floes, which sits between the open ocean and the quasi-continuous pack ice, is known as the marginal ice zone (MIZ). Following the breakup, waves herd floes [5], introduce warm water and overwash the floes, thus accelerating ice melt [6], and cause the floes to collide, which erodes the floes and influences the large-scale deformation of the ice field via momentum transfer [7]. Waves, therefore, have a substantial role in controlling the ice extent.

Accounts of wave-induced breakup in the literature are serendipitous and rarely accompanied by data, due to the difficulties in making measurements in the harsh and dynamic MIZ conditions. Models of ice breakup, which are based on strains imposed by waves exceeding a specified failure strain, are being embedded into operational forecasting models (see, for example, [8]). This includes the crucial process of wave energy attenuation over distance due to the presence of ice cover. Attenuation dictates the spatial distribution of wave energy in the ice-covered ocean, and hence the region susceptible to breakup.

A series of pioneering experiments were conducted in the Arctic Ocean in the 1970s to early 1980s [2]. The experiments were a catalyst for development of mathematical attenuation models. The models are conventionally based on an accumulation of scattering events [9] and/or parameterised viscous dissipation [10]. Notably, they are also based on linear theories, i.e. the attenuation rate is independent of wave amplitude. Field observations [4] confirm a linear regime exists for attenuation of small-amplitude waves. However, there is a transition in attenuation rates as wave amplitude increases (this is substantiated by a number of field and laboratory experiments, see e.g. [11–13]). Thus, the linear model is inaccurate for large-amplitude waves [14], and, consequently, erroneously predicts wave-induced breakup by up to hundreds of kilometres.

Field experiments are dominated by the natural environment and thus no control is possible, leaving a number of open question in the interpretation of observations. Laboratory experiments, on the other hand, offer a controlled environment. But, previous tests have been limited to wave attenuation as induced by plastic or wooden plates [12–14]. Here we present an experimental model for wave attenuation due to a continuos ice cover in the new ice-wave facility at the University of Melbourne. Wave attenuation rates, ice breakup and wave propagation over compact and broken ice is discussed for a range of different incident monochromatic waves.

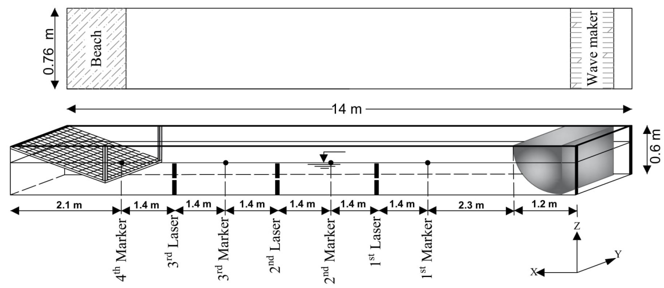


Figure 1: THE WAVE-ICE FLUME.

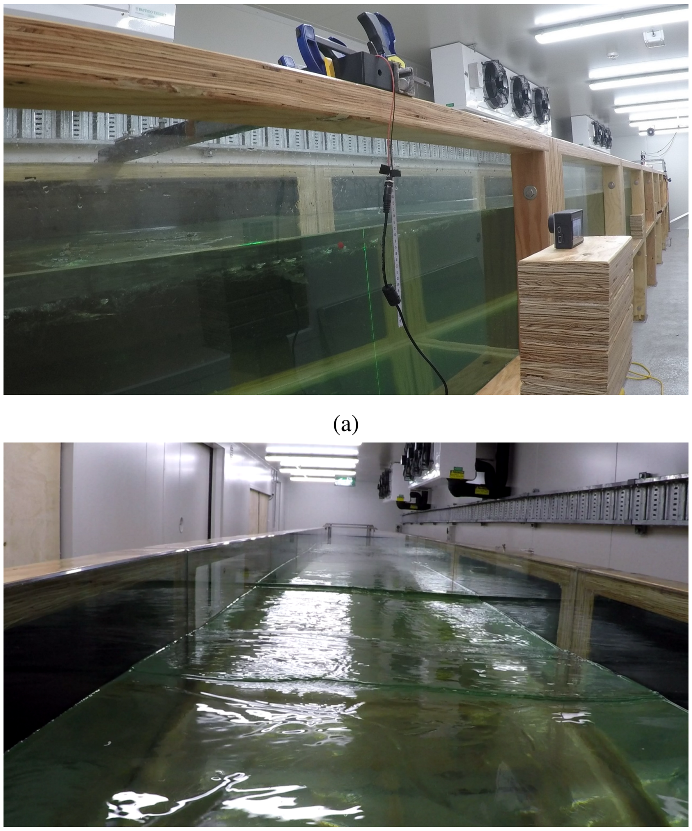
EXPERIMENTAL SETUP AND METHODOLOGY

The facility consists of a flume of wooden frame and glass walls and bottom. The flume is mounted in a refrigerated chamber, where the temperature can be lowered down to -12°C. The length of the flume is 14 m and the width is 0.76 m. For the present experiment, the flume was filled with 0.45 m of fresh water. The flume is equipped with a wave maker on one end and a linear beach of 1:6 slope at the opposite end. A preliminary analysis of the reflection, indicated that more than 95% of the incoming wave energy is attenuated by the beach. A schematic of the flume is shown in Fig. 1.

The initial condition is the still water surface at ambient temperature. The process starts by lowering the air temperature to -1°C. This condition is kept for 24 hours to ensure that the water temperature cools down to 0–1°C. At this stage, the freezing process begins by dropping the air temperature to -12°C. This condition is maintained for 5 hours. As the ice sticks at the side wall, the air temperature is then raised to -1 °C to let ice detach and thus obtaining a continuous floating ice cover. Ice thickness varied slightly during each test. Overall, it was recorded to be 0.01 ± 0.002 m. Nearly two metres of ice cover in front of the wave maker were eliminated before starting the experiment to allow the initial wave propagation to be undisturbed.

Mechanical properties of the ice were analysed with a set of axial and three-point flexural tests. Cylindrical prisms with ratio of height to diameter equal to 2 of fresh water ice were produced for axial load testing specifically, while ice cover samples (length of about 400 mm) for the three-point flexural test were taken in the flume after breaking the continuos ice cover. Five ice samples were used for each test. Results were used to compute the average Young's modulus of the fresh water ice. On average, our fresh water ice was characterised by Young's modulus of 5.3 ± 1.4 GPa. This is higher than the Young's modulus of sea ice, meaning that ice in the flume is more rigid that its sea counterpart.

The main purpose of the experiment is tracking the evolution of regular waves along the ice cover to infer wave attenuation rates over distance. Wave propagation along continuous ice



(a)

(b)

Figure 2: (a): THE RED MARKER AND LASER BEAM; (b): ICE COVER BREAKS DURING THE TEST.

is monitored by video tracking (from the side window) the vertical displacements of markers embedded in the ice cover at four distances from the wave maker: $x = 2.3$, 5.1, 7.9, and 10.7 m. Wave motion in broken ice, on the other hand, is monitored by tracking the length of a vertical, downlooking laser beam at three different locations along the flume (at 3.7, 6.5, and 9.3 m from the wavemaker). Fig. 2 shows the red marker, laser beam, camera layout and the provided continuous ice cover breaking as test starts.

Regular incident waves of different characteristics were tested. Wave period was set equal to 0.8, 1 and 1.2 s corresponding to wavelength of 1, 1.56 and 2.1 m. The wave amplitude was chosen to reach a desired wave steepness of ka, where k is the wavenumber associated to the wave period and a is the the wave amplitude, equal to 0.04 and 0.08. These conditions range between a very gently sloping wave to storm-like conditions, respectively. For each wave configuration, ice behaviour was monitored for a total time frame of 100s.

WAVE ATTENUATION

Unbroken (Continuous) Ice Cover

For each test, the videos have been analyzed to extract the vertical displacements $\eta(t)$ of the markers. Note that the displacement includes the sinusoidal oscillation due to the incoming wave and a downward shift imposed by waves running on ice (this is normally referred to as overwash [12, 14]). Fig. 3 presents the recorded displacements along the flume for an incoming wave of ka = 0.08 and T = 0.8 s (L = 1m). Close to the wave maker (x = 2.3 m), the ice cover starts oscillating as waves approach. Due to the small freeboard (a few millimetres), waves overwash the ice. This pushes the ice underwater as recorded by the downward trend of the marker's displacements. Melting is accelerated by the submersion and the ice cover eventually breaks up. The hydrostatic pressure pushes the broken, smaller, ice floes upwards, justifying the upward trend of the displacements. The floes remain submerged, though. As waves penetrate deep into the ice cover, the wave energy is attenuated, weakening the effect on the ice cover. Already at x = 5.1 m from the wave maker, downward or upward trend do not occur as overwash and breakup were not recorded (see panel b in Fig. 3). It is interesting to note, however, that the mean oscillation value seems to raise after about 50 s. Visual observations suggest that such upwelling is driven by broken ice originated at upstream sections. For farther markes, neither overwash or break up or related effects were detected. Farther from the wave maker, wave energy was substantially diminished.

Overall, break up remained confined at the very beginning of the ice cover for wave configurations of small steepness (ka = 0.04) and short periods (T = 0.8 s). For increasing wave periods and energy, break up extends toward the middle and even end of the flume. An example of floe length distribution recorded in the flume after a total break up is presented in Fig. 4 for a wave configuration of ka = 0.08 and T = 1.2 s (L = 2.2 m). Floes are rather small (1/4 of the wave length). Floe length increases irregularly moving toward the end of the basin. In the middle of the tank, floes were about half the wavelengths, while the ice cover remained unbroken at the end of the flume). It is worth mentioning that the wave period was observed to play a major role on the break up; regardless of the steepness, wave period longer than 1 s induced the break up. To a certain extent this is not unexpected as the longer period increases the bending moments of the ice (for a fixed value of steepness).

In order to evaluate the rate of attenuation of the wave field, an average wave height was extracted with standard zero-crossing method. For consistency, only a short time window, a few tens of seconds, at the start of the tests were used to ensure that average wave height is not contaminated by not only overwash but also break up effecting on the beginning of the cover. For convenience, the average wave height H is normalised by the incident wave height H_o. Fig. 5 shows the variation of the average wave height as a function of distance for representative val-

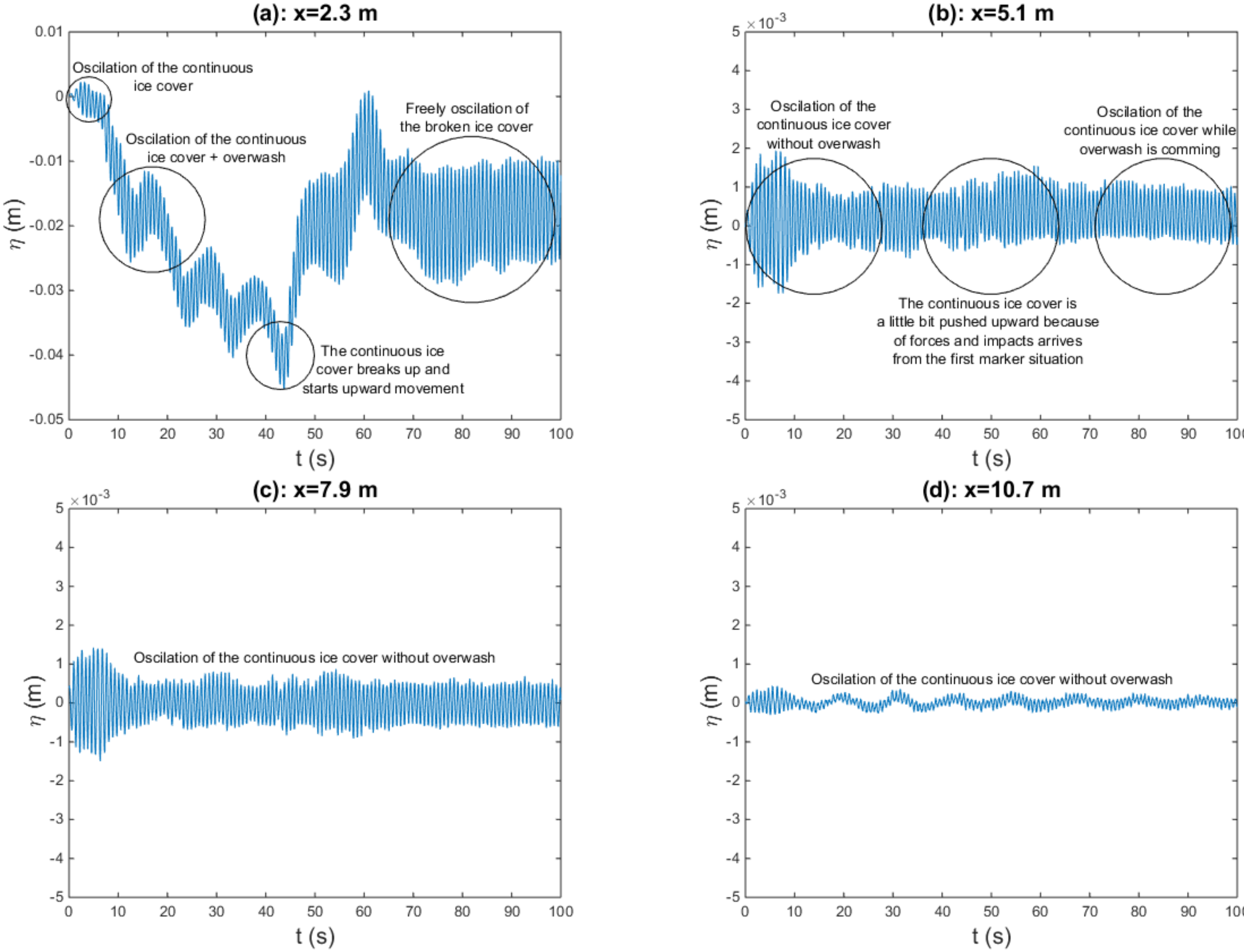


Figure 3: EXAMPLE OF OSCILLATION OF THE ICE COVER UNDER THE EFFECT OF WAVES WITH KA=0.08 & T=0.8 S.

ues of ka and T. Overall, the attenuation is exponential in agreement with field observations [15]. It is interesting to note, however, that the majority of energy attenuates at the beginning of the ice cover for very gently sloping waves. For large steepness (ka = 0.08, lower panel in Figs. 5), the attenuation rate is more substantial with only 30% of the energy propagating through the first marker.

Broken Ice Cover

In a configuration of broken ice, waves were observed to behave differently compared with a condition of compact ice. This is primarily related to the fact the broken ice floes move and collide under the effect of the waves, concurrently affecting the wave motion (especially at farther section from the wave maker). Because of the mobility of the floes, the markers were not trackable and the oscillations were monitored by a down-looking laser beam (length of the visible beam into the water was measured using image analysis). Same incident waves as the continuous ice cover case generated through this broken ice cover which consequently attenuated incident waves (L = 2.1 m). Fig. 6 shows the variation of average wave height as a function of distance for representative values of ka and T. Again the attenuation rate is exponential. For very gently sloping waves (ka=0.04) and short periods (T=0.8), attenuation is basically immediate. For longer

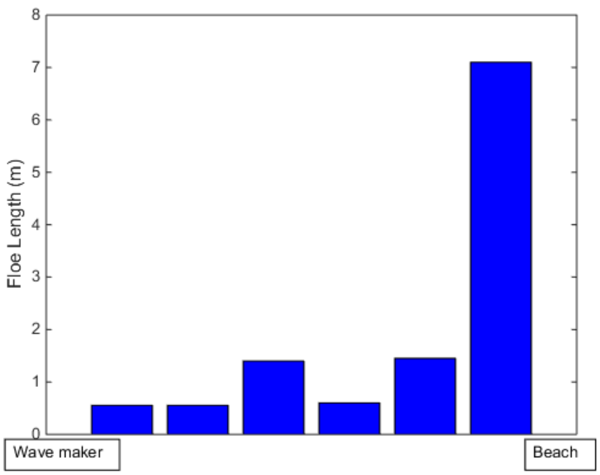


Figure 4: BROKEN ICE COVER LENGTHS UNDER STEEPNESS AND PERIOD OF 0.08 AND 1.2 S.

periods, waves penetrate the broken ice cover more easily than in a condition of compact ice (at a few wavelengths from the wave maker $H/H_0 \approx 0.5$ in broken ice and ≈ 0.3 in compact ice).

Interestingly enough, same behaviour is observed for high steepness, with energy dissipating slower than in a compact ice configuration already at the beginning of the ice cover (at a few wavelengths from the wave maker $H/H_0 \approx 0.4$ in broken ice and ≈ 0.2 in compact ice).

CONCLUSION

A study has been conducted in the new sea-ice-wave laboratory facilities of the University of Melbourne to investigate the attenuation of monochromatic waves propagating through a continuous ice cover. The facility consists of a 14 m long and 0.76 m wide wave flume (water depth of 0.45m) in a refrigerated chamber, where air temperature can be lowered down to -12°C. A continuous ice cover of thickness about 0.01 m was generated by freezing the water for 5 hours. Sinusoidal waves were generated mechanically with different periods (0.8, 1 and 1.2 s) and wave amplitudes (steepness ka = 0.04 and 0.08). The evolution of the wave field was monitored by tracing the displacement of markers embedded in the compact ice and a laser beam in a configuration of broken ice.

Results show that the attenuation rates of waves propagating through ice covers follow an exponential pattern. They also show it is a function of wave period and steepness so that increasing the wave period and steepness raise the attenuation rate of waves. A substantial attenuation rate occurs within a short distance from the wave maker, and then decreases until it goes to zero at the far end. Broken ice covers also attenuate waves but less than the continuous ice cover. To some extent, however, a proportion of the wave attenuation is related to an increase of overwash activity [13, 14, 16]. This took place not only in the presence of small floes of broken ice, but also at the begining of the compact ice where it contributes to broken up the cover at gently sloping waves.

The experiment, however, is still a strong approximation of reality and results should be interpreted with care, before they can be applied in the field.

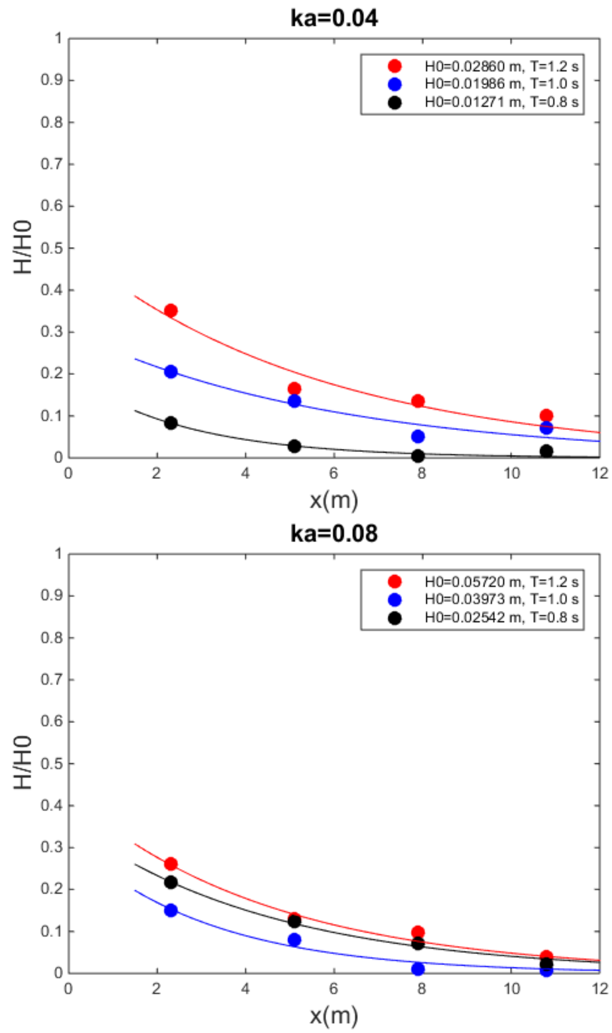


Figure 5: WAVE HEIGHT ATTENUATION RATE OF CONTINUOUSE ICE COVER ALONG THE FLUME WHEN KA=0.04 AND 0.08.

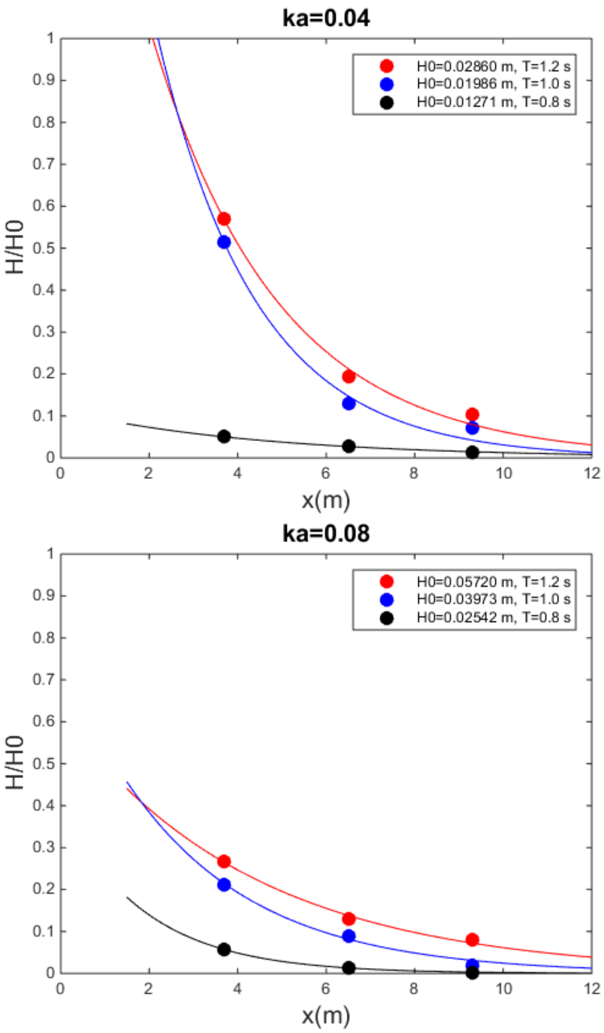


Figure 6: WAVE HEIGHT ATTENUATION RATE OF BROKEN ICE COVER ALONG THE FLUME WHEN KA=0.04 AND 0.08.

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